Professional Practices (CSC4102)

Lecture 1

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Books and Helping Materials

Text Book:

Professional Issues in Software Engineering by Frank Bott, Allison Coleman, Jack Eaton and Diane Rowland, CRC Press; 3rd Edition (2000). ISBN-10: 0748409513

Reference Book:

Computer Ethics by Deborah G. Johnson, Pearson; 4th Edition (January 3, 2009). ISBN-10: 0131112414.

Course Outline

1. Computing Profession
2. Computing Ethics
3. Philosophy of Ethics
4. The Structure of Organizations, Finance and Accounting, Anatomy of a Software House, Computer Contracts, Intellectual Property Rights
5. The Framework of Employee Relations Law and Changing Management Practices
6. Human Resource Management and IT
7. Health and Safety at Work
8. Software Liability
9. Liability and Practice
10. Computer Misuse and the Criminal Law
11. Regulation and Control of Personal Information
12. Overview of the British Computer Society Code of Conduct
13. IEEE Code of Ethics, ACM Code of Ethics and Professional Conduct
14. ACM/IEEE Software Engineering Code of Ethics and Professional Practice
15. Accountability and Auditing
16. Social Application of Ethics

Marks Distribution

Marks Distribution (Spring 2021)			
Marks Head	**Total Frequency**	**Marks /Frequency**	**Total Marks /Head**
Quiz	3	5	15
Assignment	3	5	15
Final Paper	1	40	30+10=40
Mid Term Paper	1	30	25+5=30
		Total Marks	**100**

Computing Ethics

Ethics?

- What is it?
- What does it have to do with computing?
- Why are we discussing it?

What is Ethics?

- Ethics: The philosophy of morality.
  - Ethics deals with right and wrong behavior. . .
  - But who decides what is right or wrong?

Societal Ethical Codes

- **Ancient Babylonia:** "An eye for an eye."
- **Ancient Hebrew law:** "Love thy neighbor as thyself."
- **Ancient India (circa 1500 BC):** Honesty, rectitude, charity, nonviolence, modesty, and purity of heart are virtues . .

Societal Ethical Codes

- **Ancient China (Confucius):** “What you do not want done to yourself, do not do to others.
- **Christianity:** “Do unto others as you would have them do unto you.”
- **Roman (Marcus Aurelius):** There is a universal moral law; moral relativism is rejected.

Ethical Theories

- **Relativism:**
  - There is no absolute right or wrong.
  - Ethical **relativism** is the **theory** that holds that morality is relative to the norms of one's culture. That is, whether an action is right or wrong depends on the moral norms of the society in which it is practiced

Ethical Theories

- **Universalism:**
  - **Consequentialism**
    - Does the consequence of a certain behavior favor the common good?
  - **Deontologism:**
    - Is the behavior itself good or bad?
    - The morality of an action should be based on whether that action itself is right or wrong under a series of rules, rather than based on the consequences of the action.

What does it have to do with Computing?

- **Ethics:** “The rules or standards governing the conduct of the members of a profession.”
  - [The American Heritage Dictionary, 1983]

Professions That Have Ethical Codes

- Medicine
- Law
- Engineering
- Computing
- Others. . . Professions that have the potential for great harm.

Common Themes of Computer Society Codes

- Dignity and worth of other people
- Personal integrity and honesty
- Responsibility for work
- Confidentiality of information
- Public safety, health, and welfare
- Participation in professional societies to improve the profession
- Knowledge and access to technology = social power

Ethical Codes in Computing

Ethical Codes in Computing

Ethical codes in computing are formal guidelines that define the moral principles and professional standards that computing professionals must follow. These codes help ensure responsible use of technology and protect individuals, organizations, and society from harm.

As computing plays a major role in areas like banking, healthcare, education, AI, and cybersecurity, ethical behavior is critical.

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1. Why Ethical Codes Are Important

- Protect user privacy and data
- Prevent misuse of technology
- Promote fairness and transparency
- Maintain public trust in computing systems
- Guide professionals in difficult decisions
- Reduce legal and professional risks

Ethical Codes in Computing

2. Major Ethical Codes in Computing

(A) ACM Code of Ethics (Association for Computing Machinery)

The ACM Code is one of the most widely recognized ethical standards.

Key Principles:

1. Contribute to society and human well-being
2. Avoid harm
3. Be honest and trustworthy
4. Respect privacy
5. Honor confidentiality
6. Give proper credit for intellectual property
7. Maintain professional competence

(B) IEEE Code of Ethics

The IEEE code applies to engineers and computing professionals.

Key Principles:

1. Prioritize public safety and welfare
2. Avoid conflicts of interest
3. Be honest and realistic in claims
4. Reject bribery
5. Improve understanding of technology
6. Treat all persons fairly

Ethical Codes in Computing

3. Core Ethical Principles in Computing

1. Privacy

Protecting personal and sensitive information from unauthorized access.

Example:
A software developer must not sell user data without consent.

2. Confidentiality

Keeping entrusted information secure.

Example:
A database administrator must not disclose company secrets.

3. Integrity

Ensuring accuracy and reliability of systems and data.

Example:
A programmer should not manipulate test results.

4. Intellectual Property

Respecting copyrights, patents, and licenses.

Example:
Not copying software illegally.

5. Accountability

Taking responsibility for one's actions and system outcomes.

6. Fairness and Non-Discrimination

Avoiding biased algorithms or discriminatory systems.

Example:
AI systems should not unfairly reject candidates based on gender or ethnicity.

Ethical Codes in Computing

4. Common Ethical Issues in Computing

- Data breaches
- Hacking and cybercrime
- Software piracy
- AI bias
- Surveillance misuse
- Plagiarism
- Fake information generation

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5. Ethical Decision-Making Process

When facing an ethical dilemma:

1. Identify the problem
2. Determine who is affected
3. Refer to professional codes
4. Consider possible consequences
5. Choose the action that minimizes harm and maximizes fairness

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6. Conclusion

Ethical codes in computing provide a framework for responsible professional behavior. As technology continues to grow—especially in AI, IoT, and data systems—following ethical standards ensures that computing benefits society rather than harming it.

Ethical Codes in Medicine

Ethical Codes in Medicine

Ethical codes in medicine are formal principles and guidelines that govern the conduct of doctors and other healthcare professionals. Their main purpose is to protect patients, ensure professional integrity, and promote trust in the healthcare system.

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1. Why Ethical Codes Are Important in Medicine

- Protect patient rights and dignity
- Ensure safe and fair treatment
- Promote trust between doctor and patient
- Guide doctors in difficult moral decisions
- Maintain professional standards

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2. Major Ethical Codes in Medicine

(A) Hippocratic Oath

One of the oldest ethical codes in medicine (originating in ancient Greece).

Main ideas:

- "Do no harm" (Non-maleficence)
- Act in the patient's best interest
- Maintain patient confidentiality
- Teach medicine ethically to the next generation

Ethical Codes in Medicine

(B) World Medical Association (WMA) – Declaration of Geneva

Modern version of the Hippocratic Oath.

Key principles:

- Respect patient autonomy and dignity
- Practice with conscience and integrity
- Maintain patient confidentiality
- Not discriminate based on race, religion, gender, or status

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(C) American Medical Association (AMA) Code of Medical Ethics

Provides detailed guidance on:

- Patient-physician relationship
- Informed consent
- End-of-life care
- Confidentiality
- Professional conduct

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3. Core Ethical Principles in Medicine

Medical ethics is commonly based on four fundamental principles:

1. Autonomy

Respecting the patient's right to make their own decisions.

Example:
A patient has the right to refuse treatment.

Ethical Codes in Medicine

2. Beneficence

Acting in the best interest of the patient.

Example:

Recommending the most effective treatment.

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3. Non-Maleficence

"Do no harm."

Example:

Avoiding treatments where risks outweigh benefits.

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4. Justice

Treating patients fairly and equally.

Example:

Providing equal care regardless of social status.

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4. Common Ethical Issues in Medicine

- Confidentiality breaches
- Informed consent problems
- End-of-life decisions (euthanasia)
- Organ transplantation ethics
- Medical negligence
- Allocation of limited resources (e.g., ICU beds)
- Genetic testing and privacy

Ethical Codes in Medicine

5. Ethical Decision-Making in Medicine

When facing an ethical dilemma, doctors should:

1. Identify the ethical issue
2. Consider patient rights and medical principles
3. Review professional codes and laws
4. Consult colleagues or ethics committees
5. Choose the action that best protects patient welfare

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6. Conclusion

Ethical codes in medicine ensure that healthcare professionals provide safe, fair, and respectful treatment. They maintain trust between doctors and patients and help guide medical decisions in complex situations.

Computer-related Offenses

- Offenses that existed before computers but are facilitated by computers
- Offenses against computer hardware and software
- Invasion of privacy

Scenario 1

A programmer at a bank realized that he had accidentally overdrawn his checking account. He made a small adjustment in the bank's accounting system so that his account would not have the additional service charge assessed. As soon as he deposited funds that made his balance positive again, he corrected the bank's accounting system.

Scenario 1

A programmer at a bank realized that he had accidentally overdrawn his checking account. He made a small adjustment in the bank's accounting system so that his account would not have the additional service charge assessed. As soon as he deposited funds that made his balance positive again, he corrected the bank's accounting system.

The programmer's modification of the accounting system was:

Unethical: I chose unethical because he used his position as a programmer to override the system to make adjustment to his bank account system. He abused his abilities to mess with a banking system. If anyone else would have done it they would have been tried in a court of law and more than likely jailed. Now I didn’t put it as Very Unethical because he did go back and corrected his wrong doing.

Case Study # 1

- As a computer programmer working at a bank, you discover that you have accidentally been given write access to payroll data for all bank employees. Do you give yourself a raise?

Case Study # 2

- As a computer programmer working at a bank, you discover that you have accidentally been given read access to payroll data for all bank employees. Do you compare your pay to that of other programmers?

Case Study # 3

- As a database administrator, you have been given read access to personnel data. Do you search for the private home phone number of the person you would most like to get to know?

Case Study # 4

- Your nerdy roommate has carelessly left his password where you can find it. Do you threaten anyone using his account?

Case Study # 5

- Your best friend offers to let you copy the latest version of Windows. Do you?

Case Study # 6

- Consider this statement of Mr. Jonah Nonimous: “I’m a law-abiding citizen, I pay my taxes promptly. I have a loving family. I don’t care if anyone reviews my college grades or income tax records, because I have nothing to hide. All these privacy laws are unnecessary. Only individuals who have something to hide need them.”

Why are we discussing this?

- Because you are working toward becoming a computer professional who will have knowledge that many do not, who will have access to information that many do not, who will have the ability to cause harm or to do good.

Organizations of Computing Professionals That Have Ethical Codes

- Association of Information Technology Professionals (AITP)
- Institute for the Certification of Computer Professionals (ICCP)
- Association for Computing Machinery (ACM)
- Institute for Electrical and Electronics Engineers (IEEE) Computer Society
- British Computer Society (BCS)
- Canadian Information Processing Society (CIPS)
- Independent Computer Consultants Association (ICCA)

Links to Organizations of Computing Professionals That Have Ethical Codes

- **AITP** http://www.aitp.org/index.jsp
- **ICCP** http://www.iccp.org/iccpnew/ethics%20practice%20conduct.html
- **ACM** http://www.acm.org/constitution/code.html
- **IEEE** http://www.ieee.org/about/whatis/code.html
- **BCS** http://www.bcs.org/
- **CIPS** http://www.cips.ca/about/ethics/
- **ICCA** http://www.icca.org/ethics.asp